

3.5.3 Cleaning Solution Preparation and Cleaning Procedure

The concentration and necessary quantity of each chemical are shown in Table 3.1. If the delivered concentration is higher than shown below, dilute it in the feed tank with clean water.

Chemical	Solution concentration	Amount used
Sodium hypochlorite	2,000-6,000 mg/l (Effective chlorine concentration) (pH is about 12) Preventive: 1,000-2,000 mg/l	5 l/element (TMR140) or 3 l/element (TMR090)
Oxalic acid	5-10 g/l	
Citric acid	10-30 g/l Preventive: 2-5 g/l	

Table 3.1 Chemical preparation guideline

Cleaning procedure (see above Figure 3.1):

1. Measure permeability (flux rate, TMP and liquid temperature) before starting chemical cleaning.
2. Check that the chemical injection valve is closed and that the chemical feed pump is at rest.
3. Feed the feed tank with specified amount of chemical and dilute if necessary.
4. Stop filtration and air diffusion and close the permeated water valve.
5. Start the chemical feed pump and check overflow from head tank.
6. Slowly open the chemical injection valve to inject the chemicals. Adjust the injection rate by controlling the valve to achieve recommended injection time (around 30 minutes). A fast injection may lead to insufficient fulfillment of chemical to the elements.
7. After injecting the specified amount of chemicals, stop the chemical feed pump. If specified amount cannot be injected, check the injection rate and injection piping. NEVER force to inject surplus chemical which may result in severe damage to the membrane.
8. Leave the equipment for 1 to 3 hours.
9. Close the chemical injection valve, open the permeated water valve and restart filtration. Chemicals remain in the permeated water in an early phase of filtration (for a period of 2 or more intermittent cycles). If it is inconvenient for the plant operation, send the permeated water back to the raw water tank or dispose of it in accordance with applicable legal standards for waste disposal.

Regarding permeate piping cleaning, just add a small amount of sodium hypochlorite solution while filtrating. Note the remark mentioned in above #9.

3.5.4 Chemical Feeding Method for Multiple Modules

When several modules are connected to a permeate header, it is possible that chemical will be injected to the header pipe, although it is preferable to inject directly into manifold of each module by chemical header. To achieve equal distribution to all modules, the following procedure is recommended:

- Install one injection point to the header pipe per every ten (10) modules connected to the pipe and inject the chemical evenly from each point.
- When the header pipe is inclined, injection should be done more from the higher point than the lower point.

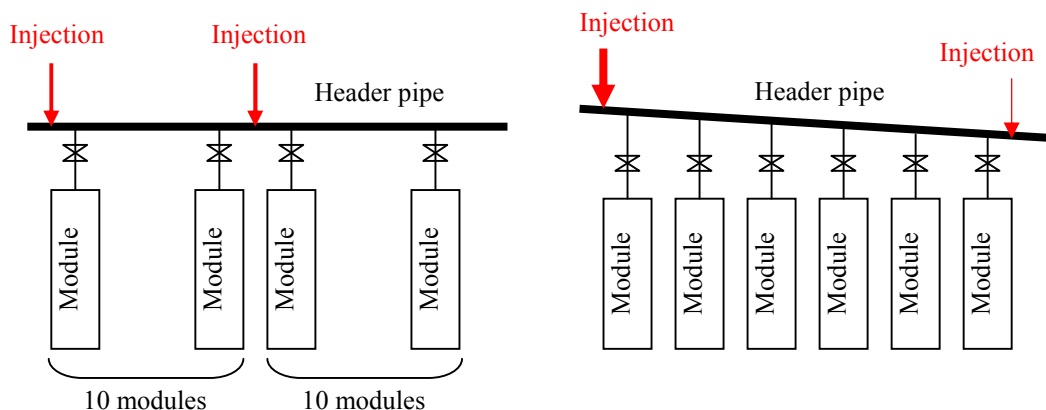


Figure 3.2 Chemical feeding points



WARNING!

If an abnormality is found in the equipment during chemical cleaning, immediately stop the operation.



WARNING!

If chemicals are injected forcibly with the chemical feed pump or by any other means, the internal pressure of the element may increase, leading to damage to the element. Be sure to inject chemicals by gravity at 10 kPa or less.



WARNING!

Before feeding chemicals for chemical cleaning, check that the water surface is 500 mm or more above the top of the module. Feed chemicals only after "MEMBRAY" is completely submerged.



DANGER!

Toxic chlorine vapors will be produced if acid and hypochlorite are mixed together. Always ensure that one chemical is completely rinsed from the tank and piping prior to cleaning with the other chemical. It is recommended that, after the chemical make up tank is drained, it will be rinsed it thoroughly, refilled with clean water and be flushed through all contaminated piping until residual chlorine levels are less than 10 ppm. If chlorine vapors are detected, the operator should evacuate the area immediately and advise the plant safety officer and / or immediate supervisor accordingly.

Notes:

If permeate is used as cleaning solution make up water, residual COD and ammonia in permeate will consume chlorine used for standard chemical cleaning

3.5.3 Cleaning Solution Preparation and Cleaning Procedure

The concentration and necessary quantity of each chemical are shown in Table 3.1. If the delivered concentration is higher than shown below, dilute it in the feed tank with clean water.

Chemical	Solution concentration	Amount used
Sodium hypochlorite	2,000-6,000 mg/l (Effective chlorine concentration) (pH is about 12) Preventive: 1,000-2,000 mg/l	5 l/element (TMR140) or 3 l/element (TMR090)
Oxalic acid	5-10 g/l	
Citric acid	10-30 g/l Preventive: 2-5 g/l	

Table 3.1 Chemical preparation guideline

Cleaning procedure (see above Figure 3.1):

1. Measure permeability (flux rate, TMP and liquid temperature) before starting chemical cleaning.
2. Check that the chemical injection valve is closed and that the chemical feed pump is at rest.
3. Feed the feed tank with specified amount of chemical and dilute if necessary.
4. Stop filtration and air diffusion and close the permeated water valve.
5. Start the chemical feed pump and check overflow from head tank.
6. Slowly open the chemical injection valve to inject the chemicals. Adjust the injection rate by controlling the valve to achieve recommended injection time (around 30 minutes). A fast injection may lead to insufficient fulfillment of chemical to the elements.
7. After injecting the specified amount of chemicals, stop the chemical feed pump. If specified amount cannot be injected, check the injection rate and injection piping. NEVER force to inject surplus chemical which may result in severe damage to the membrane.
8. Leave the equipment for 1 to 3 hours.
9. Close the chemical injection valve, open the permeated water valve and restart filtration. Chemicals remain in the permeated water in an early phase of filtration (for a period of 2 or more intermittent cycles). If it is inconvenient for the plant operation, send the permeated water back to the raw water tank or dispose of it in accordance with applicable legal standards for waste disposal.

Regarding permeate piping cleaning, just add a small amount of sodium hypochlorite solution while filtrating. Note the remark mentioned in above #9.

3.5.4 Chemical Feeding Method for Multiple Modules

When several modules are connected to a permeate header, it is possible that chemical will be injected to the header pipe, although it is preferable to inject directly into manifold of each module by chemical header. To achieve equal distribution to all modules, the following procedure is recommended:

- Install one injection point to the header pipe per every ten (10) modules connected to the pipe and inject the chemical evenly from each point.
- When the header pipe is inclined, injection should be done more from the higher point than the lower point.

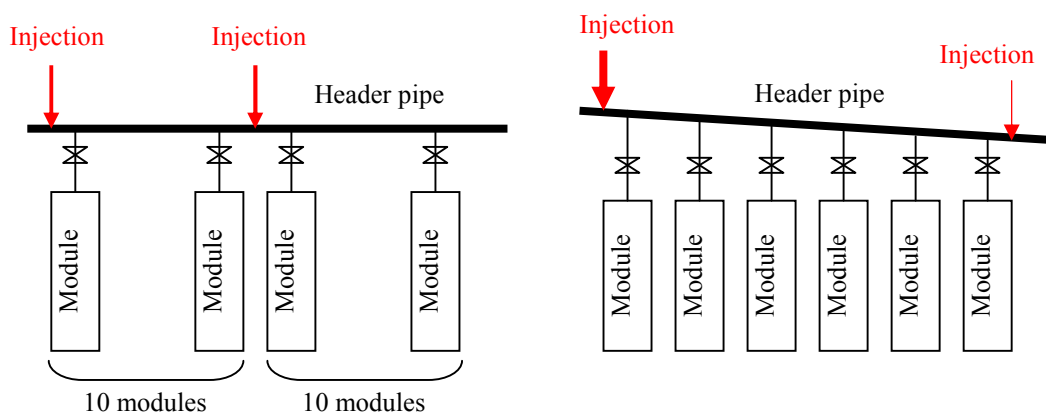


Figure 3.2 Chemical feeding points



WARNING!

If an abnormality is found in the equipment during chemical cleaning, immediately stop the operation.



WARNING!

If chemicals are injected forcibly with the chemical feed pump or by any other means, the internal pressure of the element may increase, leading to damage to the element. Be sure to inject chemicals by gravity at 10 kPa or less.



WARNING!

Before feeding chemicals for chemical cleaning, check that the water surface is 500 mm or more above the top of the module. Feed chemicals only after "MEMBRAY" is completely submerged.



DANGER!

Toxic chlorine vapors will be produced if acid and hypochlorite are mixed together. Always ensure that one chemical is completely rinsed from the tank and piping prior to cleaning with the other chemical. It is recommended that, after the chemical make up tank is drained, it will be rinsed thoroughly, refilled with clean water and be flushed through all contaminated piping until residual chlorine levels are less than 10 ppm. If chlorine vapors are detected, the operator should evacuate the area immediately and advise the plant safety officer and / or immediate supervisor accordingly.

Notes:

If permeate is used as cleaning solution make up water, residual COD and ammonia in permeate will consume chlorine used for standard chemical cleaning

(approximately 1 ppm COD/ammonia consumes several ppm of chlorine). Measure the chlorine concentration in the standard chemical cleaning solution periodically to ensure the desired concentration is achieved. If the permeate COD is high (typical in industrial plants) and if it reduces the chlorine in the solution, use tap water for standard chemical cleaning or dose extra NaOCl.

Also, residual calcium in the make up water may precipitate when NaOCl is mixed into the water. This may cause scaling and fouling of the membrane. If a high calcium concentration is expected in the make up water, confirm the possibility of precipitation by a bench-scale NaOCl addition test and if whitish precipitation of calcium is found, use tap water for the make up water.

Additional Remarks:

- Foulant in wastewater treatment is normally a mixture of different types of chemicals (organics and inorganics). While chlorine based cleaner (NaOCl) is very effective for removing biological and organic foulants, other chemical cleaners such as acids may be required for removing inorganic foulant.
- Rotational standard chemical cleaning with NaOCl and acid is recommended: Perform one standard chemical cleaning with NaOCl and the other with acid.
- Extent of cleaning with each chemical depends on the nature of predominant foulants (organic or inorganic).
- It is recommended to select the best cleaning chemical(s), concentration and frequency for each site either during pilot tests or on full-scale plants during the initial operating period.

3.6 Intensive Chemical Cleaning (after operational emergencies)

Operational emergencies or inadequate standard chemical cleaning may require the need for an intensive chemical cleaning. Intensive chemical cleaning is a procedure that can be used to restore the membrane performance to its nearly original state. Membrane permeability is measured before and after cleaning in sludge and in clean water.

The objective of intensive chemical cleaning is to achieve:

- Clean water permeability after cleaning of 75 - 85% of the new membrane clean water permeability or
- Permeability in sludge after cleaning that is at least 50% higher than the operating permeability in sludge immediately before the cleaning.

For intensive chemical cleaning, the same chemical cleaners as for standard chemical cleaning can be used but at lower concentrations and longer contact time. While standard chemical cleaning is regularly done (if TMP increases >50 mbar) intensive chemical cleaning is TORAY's recommendation as a countermeasure after operational emergencies which results in:

- Low permeability (<50% of initial membrane permeability)
- High TMP during peak flux (>50 mbar)

The fouled module(s) is removed from the MBR tank and placed in a soak tank filled with cleaning solution. Modules are inspected and accumulated solids on the membrane surface are washed off before soaking.

3.6.1 Soak Tank Intensive Chemical Cleaning

Soak tank cleaning procedures are given below along with recommended values for the cleaning variables and options to enhance cleaning efficiency.

Basically the MBR module is removed from the MBR tank, hosed down to remove excess solids and then soaked in the cleaning solution in a separate tank for a specified time. If sludge in the MBR tank can be completely removed, the tank may be used as soak tank as long as its inner surface is chemical-resistant.

Cleaning Procedure:

1. Prepare an adequately-sized soak tank with the required equipment (suction pump/piping and aeration block/source/piping) and appropriate chemical-resistant coating.
2. Record temperature, MLSS and permeability in sludge during permeation.
3. Stop permeating from the module(s) to be cleaned.
4. Remove module and hose down to wash off sludge from the membrane (See external solids removal in section 3.4).
5. Place module(s) in the soak tank filled with cleaning solution at the desired concentration, temperature, and pH (See Table 3.3).
6. Allow membrane to soak for the designated soak period (See Table 3.3).
7. Conduct low flux filtration (<15 LMH) and aeration within operation range every hour for 5 minutes. Permeate has to be returned to soak tank to keep soaking.
8. Measure and record condition of cleaning solution.
9. Aerate the membrane for 5 minutes at the end of the soak period to scour the membrane and dislodge loose solids.
10. Return module(s) to MBR tank and re-start operation, or perform alternating cleaning (see notes below).
11. Record temperature, MLSS and permeability during permeation at the same gross fluxes as before cleaning.
12. Evaluate the cleaning efficiency based on the permeability data corrected to a common temperature (e.g. 20°C).

Parameters		Units	Values
NaOCl cleaning	Concentration	mg/l	500 - 1000
	pH		> 10
	Contact time	hr	6-24
Citric acid cleaning	Concentration	mg/l	1000 - 5000
	pH		2-3
	Contact time	hr	6-24
Liquid temperature		°C	>15

Table 3.3 Recommended values for soak tank cleaning

Notes:

The size of the soak tank should be designed so that chemical solution cannot be overflowed when immersing membrane module.

When moving the module to/from the soak tank, be careful of the trickling water from the module. Especially after soaking, this water contains NaOCl or citric acid.

Instead of returning the module to the MBR tank after cleaning with one chemical, an alternating soak cleaning can be performed with another cleaner to remove both organic and inorganic foulants (e.g., NaOCl followed by acid).

Alternating soak cleaning can be performed using the same tank or another soak tank. In either case, very important safety measures MUST be taken when performing alternating cleaning using chlorine and acid based cleaners (e.g., NaOCl followed by acid).

See safety note.

 DANGER!	Toxic chlorine vapors will be produced if acid and hypochlorite are mixed together. Always ensure that one chemical is completely rinsed from the tank and piping prior to cleaning with the other chemical. It is recommended that, after the chemical make up tank is drained, it will be rinsed it thoroughly, refilled with clean water and be flushed through all contaminated piping until residual chlorine levels are back to harmless concentrations. If chlorine vapors are detected, the operator should evacuate the area immediately and advise the plant safety officer and / or immediate supervisor accordingly.
--	---

Soak Tank Cleaning Recommendations:

The type of chemical cleaner, concentration, contact time depends on the type of foulants and the extent of fouling. Key recommendations for soak tank intensive chemical cleaning in MBR applications are given below:

Recommended values for soak tank intensive chemical cleaning variables are given in Table 3.3. While higher chemical concentration, longer contact time and higher temperature will increase the cleaning efficiency, a balance is needed to reduce chemical usage to shorten the cleaning time and minimize any negative impact on the membrane. The appropriate cleaning conditions are determined by starting from less severe and moving to more severe if better intensive chemical cleaning is needed. TORAY's recommendations are as follows:

- The first phase of this cleaning should be with 500 mg/l NaOCl at the warmest temperature possible (<35 °C) for 6 hr.
- Most aggressive cleaning is at 1000 mg/l NaOCl at 35 °C for 24 hr. Use this condition only if absolutely necessary. This must be done no more frequently than once a year.
- Acid soak tank cleaning to remove inorganic fouling is normally performed at 1000-2000 mg/l, pH of 2.0 and at the same temperature and contact time as the NaOCl clean. Longer contact time or higher concentrations can be used if the degree of inorganic fouling is severe.

3.6.2 Options to Enhance Soak Tank Cleaning

Soak tank intensive chemical cleaning is very effective at restoring membrane permeability when it is performed properly. There are important factors that can affect the effectiveness of soak tank cleaning. There are several options to improve the ease and efficiency of soak tank cleaning as outlined below.

Factors that can impact the effectiveness of soak tank cleaning

- Ensure soak tank is clean (free of sludge and organics) before adding cleaning solution. Organics will consume the cleaning chemicals and reduce the effectiveness of the soak cleaning.

- Ensure that the membrane is free of sludge before placing it in the cleaning solution.
- Temperature of cleaning solution is important. Higher temperature (up to 35°C) will improve the cleaning efficiency.
- Periodical permeation of chemical solution is very effective for cleaning membrane pores and inner-element portion.

Options for improving the ease and efficiency of soak tank cleaning:

- Aerate the membrane to remove attached sludge during soak cleaning.
- Install liquid heating equipment to perform soak cleaning at elevated temperature (up to 35 °C).
- Allow for permeating the cleaning solution through the membrane.
- Use separate soak tanks for NaOCl and acid cleaning to perform alternating cleaning and reuse cleaning solutions.

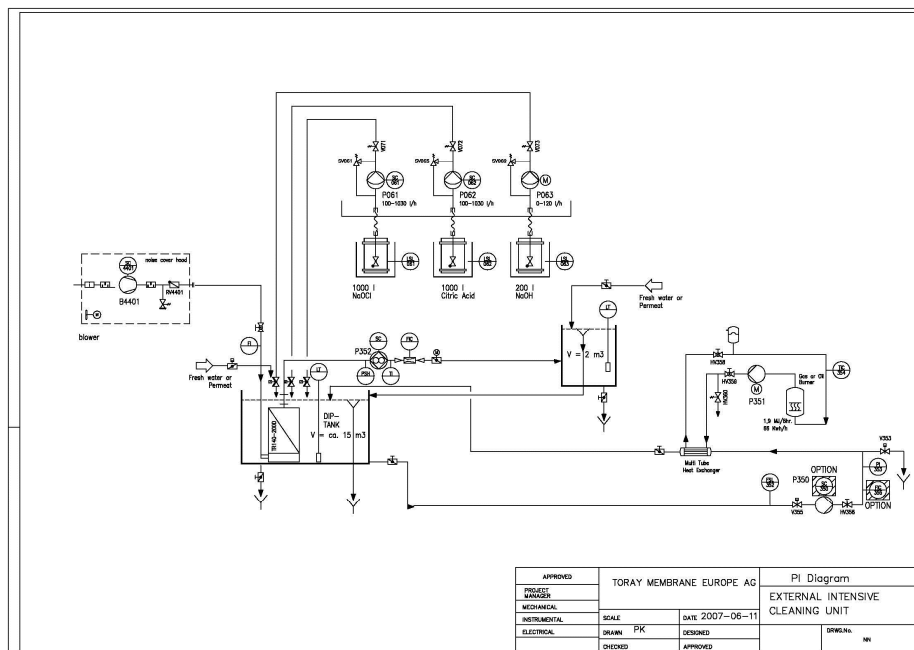


Figure 3.3 Example of soak tank cleaning facility

3.7 Calculation of Permeability

The condition of the membrane will be calculated by means of permeability. Permeability is a relative figure calculated to 1.0 m² membrane area and a theoretical differential (suction) pressure of 1.0 bar. Permeability should always be measured at the same permeate flow (flux). Depending on the condition of the membrane more or less differential (suction) pressure is required to generate a certain amount of permeate.

Example of calculation:

Prerequisites:

- | | |
|--|------------------------|
| • Number of modules of TMR140-200D per train | 3 |
| • Membrane area per module | 280 m ² |
| • Membrane area per train | 840 m ² |
| • Permeate flow of the train | 22.0 m ³ /h |

- Trans membrane pressure 0.050 bar
- Liquid temperature in MBR tank 17°C
- MLSS (for reference) 12 g/l

In this case, permeability is calculated as follows:

Calculation of permeate flux

$$Flux (l / m^2 \cdot h) = \frac{Permeate \text{ flow } (m^3 / h) \times 1000}{Membrane \text{ area } (m^2)} = \frac{22.0 \times 1000}{840} = 26.19$$

Calculation of permeability at given temperature

$$Permeability (l / m^2 \cdot h \cdot bar) = \frac{Flux (l / m^2 \cdot h)}{Trans \text{ membrane pressure } (bar)} = \frac{26.19}{0.050} = 523.8$$

Normalization of permeability to 20°C:

$$P_{T1} = P_T \times \frac{C_T}{C_{T1}}$$

$$C_T = 5.752466 \times 10^{-4} \cdot T^2 - 5.192241 \times 10^{-2} \cdot T + 1.938531$$

Where P_{T1} : Normalized permeability at $T1$ °C

P_T : Permeability at T °C

C_{T1} : Temperature correction factor at $T1$ °C

C_T : Temperature correction factor at T °C

$$C_{20} = 1.130181, C_{17} = 1.222096$$

$$P_{20} = P_{17} \times \frac{C_{17}}{C_{20}} = 523.8 \times \frac{1.222096}{1.130181} = 566.4 (l / m^2 \cdot h \cdot bar)$$

T (°C)	C_T	T (°C)	C_T
10	1.476832	25	1.000000
15	1.289125	30	0.898531
20	1.130181	35	0.825924

Table 3.4 Temperature correction factor

3.8 Handling Cleaning Chemicals

3.8.1 Avoidance of Mixture of Hypo/Acid

Depending on the application, an acid cleaner may precede or follow a hypochlorite cleaning. The system design, plant operating manual and the cleaning procedures must ensure that adequate safety protection is provided to ensure that acid and chlorine will not be mixed together.

Safety Note:

Toxic chlorine vapors will be produced if acid and hypochlorite are mixed together. Always ensure that one chemical is completely rinsed from tanks, modules and piping prior to cleaning with the other chemical.

It is recommended that the tanks must be drained completely, the tanks, piping and modules must be flushed and rinsed thoroughly with clean water.

The residual chlorine level in the rinsed water must be less than 10 ppm before cleaning with an acidic cleaner.

If chlorine vapors are detected, everyone (including the operator) must evacuate the area immediately and advise the plant safety officer and/or immediate supervisor accordingly.

There are various options available to the design engineer to reduce the risk of acid and hypochlorite mixing. These options include:

1. Provide separate tanks for preparing the chemical cleaning solutions;
2. Provide a removable fitting piece between tanks to ensure that valves cannot be left open accidentally;
3. Provide chlorine gas detectors and alarms in the area where toxic vapors could be produced;
4. Ensure in the control logic that the operator is obliged to specify if the cleaning is with acid or hypochlorite. Then provide interlocks to ensure that rinsing is required between chemical washes;
5. Install a pH probe in the chemical make up tank and interlock it to the metering pumps supplying chemical (i.e. if pH <6 chlorine pump cannot operate; if pH > 8.5 acid pump cannot operate).
6. Interlock acid & hypochlorite pumps so that they cannot operate at the same time;
7. Never allow acid cleaning to be done without an operator being present. Program the operator interface to advise the operator of the toxic vapors that are produced if acid and chlorine are mixed. Request confirmation that piping has been rinsed adequately prior to operating of the chemical metering pump(s);
8. Provide appropriate warning labels for the chemical metering tanks and pumps;
9. Ensure that the operator's manual provides adequate warnings and instructions regarding what to do in the event of vapors being produced;
10. Ensure that adequate safety equipment, in particular eye wash stations are included in the design or provided by the owner.

3.8.2 *Instruction for Each Chemical*

The chemical agents used for chemical cleaning are harmful to health when in contact with the skin. In handling chemicals, wear protective goggles, protective gloves and other protectors. Before using chemicals, be sure to check the details of its material safety data sheet (MSDS) and the instructions given below. If chemicals come into contact with skin, follow the MSDS to take actions suitable for each chemical.

Sodium hypochlorite solution / NaOCl

Handling precautions

- (a) Ventilate well. Avoid heat sources and sparks. Also avoid contact with acids.
- (b) Handle the chemical container with great care. Avoid toppling, bumping or dragging it.
- (c) Take care to prevent leak, spillover or spattering. Do not cause dust or vapor.
- (d) Firmly seal the container after use.
- (e) After using chemicals, thoroughly wash your hands and face and rinse out your mouth.

- (f) Do not eat or drink except in a designated place.
- (g) Do not bring gloves or other contaminated protectors into the rest area.
- (h) Forbid unauthorized entry to the place where chemicals are handled.
- (i) Wear appropriate protectors to avoid inhalation, eye or skin contact, and direct contact with your clothes.
- (j) To handle chemicals outdoors, provide local ventilation.

Storage precautions

- (a) Store the container in a dark, cold place. Avoid direct sunlight. Firmly seal to prevent direct contact with air.
- (b) For storage, use corrosion-resistant containers.

Oxalic acid / (COOH)₂

Handling precautions

- (a) Keep away from strong acids and bases.
- (b) Handle the chemical container with great care. Avoid toppling, bumping or dragging it.
- (c) Take care to prevent leak, spillover or spattering. Do not cause dust or vapor.
- (d) Firmly seal the container after use.
- (e) After using chemicals, thoroughly wash your hands and face and rinse out your mouth.
- (f) Do not eat or drink except in a designated place.
- (g) Do not bring gloves or other contaminated protectors into the rest area.
- (h) Forbid unauthorized entry to the place where chemicals are handled.
- (i) Wear appropriate protectors to avoid inhalation, eye or skin contact, and direct contact with your clothes.
- (j) To handle chemicals outdoors, provide local ventilation.

Storage precautions

- (a) Store the container in a dark, cold place. Avoid direct sunlight. Firmly seal to prevent direct contact with air.
- (b) For storage, use corrosion-resistant containers.

Citric acid / HOOCCH₂C(OH)(COOH)CH₂COOH

Handling precautions

- (a) Keep away from strong acids and bases.
- (b) Handle the chemical container with great care. Avoid toppling, bumping or dragging it.
- (c) Take care to prevent leak, spillover or spattering. Do not cause dust or vapor.
- (d) Firmly seal the container after use.
- (e) After using chemicals, thoroughly wash your hands and face and rinse out your mouth.
- (f) Do not eat or drink except in a designated place.
- (g) Do not bring gloves or other contaminated protectors into the rest area.
- (h) Forbid unauthorized entry to the place where chemicals are handled.
- (i) Wear appropriate protectors to avoid inhalation, eye or skin contact, and direct contact with your clothes.
- (j) To handle chemicals outdoors, provide local ventilation.

Storage precautions

- (a) Store the container in a dark, cold place. Avoid direct sunlight. Firmly seal to prevent direct contact with air.
- (b) For storage, use corrosion-resistant containers.



Some chemical agents used for chemical cleaning are harmful to the health. When ever handling chemicals, wear protective goggles, protective gloves and other protectors. Before using a chemical, be sure to check the details of its material safety data sheet (MSDS).



If chemicals stick to your skin or clothes, immediately wash it away with a large amount of running water.



If chemicals enter an eye, immediately wash it away with a large amount of running water and see the doctor.



Store chemicals in a dark, cold place free from direct sunlight.



For chemicals storage tanks, use a material suitable for each chemical to prevent corrosion.



Do not mix sodium hypochlorite with heavy metals or acids. Its mixture with an acid generates toxic chlorine gas.



Do not mix sodium hypochlorite with oxalic acid or citric acid. Such mixture generates toxic chlorine gas.